LECTURE 2

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In this lecture, we recall the construction of Lurie's spherical Witt ring and the Lubin-Tate theory attached to a perfect ring for a fixed character p, a finite height n and a given formal group $\mathbb{G}$ of height n. We combine these data into an adjunction, which is refered to as Lubin-Tate-tilt adjunction in [BSY22]. We explore the property of this adjunction, setting the stage for the next talk.

Introduction

Our main goal in this talk is to establish the following adjunction.

Theorem 0.1. *Fix a perfect field k of character p and a height n, then there is an adjunction*

$$E(-) : \mathrm{Perf}_k \rightleftharpoons \mathrm{CAlg}^{\wedge}_{E(k)} : (-)^{\flat}$$

rendering Perf_k *as a colocalization of* $\mathrm{CAlg}^{\wedge}_{E(k)}$. *Here* Perf_k *denotes the category of (ordinary) perfect k-algebras and* $\mathrm{CAlg}^{\wedge}_{E(k)} := \mathrm{CAlg}(\mathrm{Sp}_{T(n)})_{E(k)/-}$ *is the category of $T(n)$-local $E(k)$-algebras.*

The main inputs are twofold. First, we need Lurie's spherical Witt functor, which instinctively owns a right adjoint.

$$\mathbb{W}(-) : \mathrm{Perf} \rightleftharpoons \mathrm{CAlg}(\mathrm{Sp}_p) : (-)^{\flat}$$

For any p-complete presentable stable ∞-symmetric monoidal category $\mathcal{C} \in \mathrm{CMon}(\mathrm{Pr}^{\mathrm{L}}_{st})$, we have the unit functor $\iota : \mathrm{Sp} \to \mathcal{C}$ inducing the adjunction

$$\iota^* : \mathrm{CAlg}(\mathrm{Sp}_p) \rightleftharpoons \mathrm{CAlg}(\mathcal{C}) : \iota_*$$

After combining these two adjunctions and taking $\mathcal{C} = \mathrm{Mod}^{\wedge}_{E(k)}$, we will get an adjunction between $\mathbb{W}(-)$ and $(-)^{\flat}$.

$$\mathbb{W}(-) : \mathrm{Perf}_k \rightleftharpoons \mathrm{CAlg}^{\wedge}_{E(k)} : (-)^{\flat}$$

Secondly, Lubin-Tate theories allow us to functorially assign each pair $(A, n, \mathbb{G}_0)$ a $T(n)$-local $\mathbb{E}_\infty$-ring $E(A, n, \mathbb{G}_0)$. Once we fix a given pair $(k, n, \mathbb{G}_0)$ such that A takes values in perfect k-algebras, we obtain a map $E(-) : \mathrm{Perf}_k \to \mathrm{CAlg}^{\wedge}_{E(k)}$. We then compare $E(-)$ with $\mathbb{W}(-)$.

1. Thickenings and Spherical Witt rings

Recall that in classical Witt ring theory, we want to lift a ring A of character p to some universal $W(A)$ such that p is not a zero-divisor and $W(A)/pW(A) \cong A$. The Witt ring can be characterized by some universal properties. One can also construct $W(A)$ in a concrete way. For example, the underlying set of $W(A)$ consists of the formal series

$$[a_0] + [a_1]p + [a_2]p^2 + \cdots$$

and the multiplication is given by Witt polynomials.

The spherical Witt functor wants to generalize this construction to $\mathbb{E}_\infty$-settings. For example, the standard equivalence $\mathbb{Z}_p/p \xrightarrow{\cong} \mathbb{F}_p$ identifies $\mathbb{Z}_p$ with $W(\mathbb{F}_p)$. In spherical cases, we want to replace $\mathbb{Z}_p$ by $\mathbb{S}_p$. In order to make this spirit more precise, we need to introduce the concept of thickenings.

1.1. **Thickenings.**

Definition 1.1. Let A be an $\mathbb{E}_\infty$-ring, $I \in \pi_0(A)$ a finitely generated ideal, $f_0 : \pi_0(A) = A_0 \to B_0$ a map of ordinary rings. We say that $f : A \to B$ is an A-thickening of f_0 if we have the following square

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \downarrow & & \downarrow \\ A_0 & \xrightarrow{f_0} & B_0 \end{array}$$

together with the following data

(1) B is an I-complete $\mathbb{E}_\infty$-ring.
(2) The above square induces an equivalence $\pi_0(B)/I \cong B_0$.
(3) For any $\mathbb{E}_\infty$-A-algebra R, there is a natural equivalence $\mathrm{Map}_A(B, R) \simeq \mathrm{Hom}_{A_0}(B_0, \pi_0(R)/I)$.

Example 1.2. Let $A = \mathbb{Z}_p$ and B_0 be an $A_0 = \mathbb{F}_p$-algebra. Then the $\mathbb{Z}_p$-thickening of f_0 is identified with the classical Witt ring $W(A)$.

Remark 1.3. By the universal property, the thickening is unique up to equivalence when it exists.

Lurie's theorem tells us that the thickening exists under some perfect assumptions, which covers the most interesting cases.

Theorem 1.4 ([Lur17a], Theorem 5.2.5)**.** *Let A be an $\mathbb{E}_\infty$-ring, $I \in \pi_0(A)$ be a finitely generated ideal, $f_0 : A_0 \to B_0$ be a relative perfect map between perfect $\mathbb{F}_p$-algebras. Then*

(1) The A-thickening of f exists.
(2) The thickening square

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \downarrow & & \downarrow \\ A_0 & \xrightarrow{f_0} & B_0 \end{array}$$

is a pushout square of $\mathbb{E}_\infty$-rings.

In fact, in Lurie's proof, B can be constructed canonically as a filtered colimit of maping spaces, see [Lur17a, Section 5.2.1].

Example 1.5. Let $A = \mathbb{S}_p$ and $I = (p) \in A_0 = \mathbb{F}_p$. Let B_0 be a perfect $\mathbb{F}_p$-algebra. By the theorem, the $\mathbb{S}_p$-thickening of B_0 exists and we denote it by $\mathbb{W}(B)$. One can check that the construction is functorial, hence determines a functor

$$\mathbb{W}(-) : \mathrm{Perf} \to \mathrm{CAlg}(\mathrm{Sp}_p)$$

which we refer to as **spherical Witt functor**.

Before we continue to form the right adjoint, we introduce a detection for thickenings.

Proposition 1.6. *Let A be an $\mathbb{E}_\infty$-ring, $I \in \pi_0(A)$ be a finitely generated ideal, $f_0 : A_0 \to B_0$ be a relative perfect map between perfect $\mathbb{F}_p$-algebras. Then $f : A \to B$ is identified with the A-thickening of f_0 if and only if the following conditions are satisfied*

(1) B is an I-complete $\mathbb{E}_\infty$-algebra
(2) $\pi_0(B)/I \cong B_0$
(3) The thickening square

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \downarrow & & \downarrow \\ A_0 & \xrightarrow{f_0} & B_0 \end{array}$$

is a pushout square of $\mathbb{E}_\infty$-rings.

Proof. The "only if" is part of the theorem's statement, we prove the inverse. Suppose B satisfies the above data, we can identify $\pi_0(B)/I$ with B_0. By the theorem, suppose $f : A \to D$ is the A-thickening of f_0, by the universal property that $\mathrm{Map}_A(D, B) \simeq \mathrm{Hom}_{A_0}(B_0, \pi_0(B)/I)$, one can lift the identity map on B_0 to the following square

$$\begin{array}{ccccc} A & \xrightarrow{f} & D & \xrightarrow{g} & B \\ \downarrow & & \downarrow & & \downarrow \\ A_0 & \xrightarrow{f_0} & B_0 & \xrightarrow{\mathrm{id}} & B_0 \end{array}$$

The outer square is a pushout square by the third assumption and the left square is a pushout square by the theorem. Therefore, the right square is a pushout square and g is forced to be an equivalence. □

Corollary 1.7. *The essential image of $\mathbb{W}(-)$ consists of those $\mathbb{E}_\infty$-ring R such that $R \otimes_{\mathbb{S}_p} \mathbb{F}_p$ is discrete.*

We can now form the right adjoint of $\mathbb{W}(-)$.

Proposition 1.8. *We define the **tilt functor** $(-)^\flat : \mathrm{CAlg}(\mathrm{Sp}_p) \to \mathrm{Perf}$ as $(-)^\flat = (\pi_0(-)/p)^\flat$. Then $(-)^\flat$ serves as the right adjoint for $\mathbb{W}(-)$.*

Proof. By definition, one has $\mathbb{W}(A)^\flat \cong A$. We can choose the unit map as the natural equivalenceid $\xrightarrow{\simeq} \mathbb{W}(-)^\flat$. The unit map induces

$$\mathrm{Map}(\mathbb{W}(A), R) \simeq \mathrm{Map}(\pi_0\mathbb{W}(A)/p, \pi_0 R/p) \simeq \mathrm{Map}((\pi_0\mathbb{W}(A)/p)^\flat, (\pi_0 R/p)^\flat) \simeq \mathrm{Map}(A, R^\flat)$$

where the first equivalence is from definitions of $\mathbb{W}()$ and thickenings and the second is because $\pi_0\mathbb{W}(A)/p$ is perfect. □

Example 1.9 (examples of spherical Witt rings)**.** Let M be an ordinary monoid. Then the spherical Witt ring of the perfect $\mathbb{F}_p$-algebra $\mathbb{F}_p[M]$ is exactly $\mathbb{S}_p[M]$. In particular, if we take $M = \mathbb{N}[1/p]$, we have $\mathbb{W}(\mathbb{F}_p[t^{1/p^\infty}]) \simeq \mathbb{S}_p[t^{1/p^\infty}]$.

1.2. **Properties of the spherical Witt functor.** We now list some properties of the spherical Witt functor.

Lemma 1.10. *For any $\mathbb{E}_\infty$-algebra A, the spherical Witt ring $\mathbb{W}(A)$ is p-adic formally étale, i.e. the relative cotangent complex $L_{\mathbb{W}(A)/\mathbb{S}_p}$ vanishes.*

Proof. This follows from the base change formula of cotangent complex along the pushout square that defines $\mathbb{W}(A)$. □

We will not use too much about the cotangent complex. In fact, we are interested in the following consequence of formally étaleness.

Proposition 1.11 ([Lur17b]; Remark 7.4.1.8)**.** *Let R be a p-adic formally étale $\mathbb{E}_\infty$-ring, then any square-zero extension $A \to B$ induces an equivalence*

$$\mathrm{Map}(R, A) \simeq \mathrm{Map}(R, B)$$

Remark 1.12. One can check [Lur17b, Section 7] for more details about square-zero extensions and cotangent complex. We will not discuss much here. But we do concern the following two examples of square-zero extensions. First, let R be a connective $\mathbb{E}_\infty$-algebra, each connection map in the Postnikov tower of R

$$\cdots \to \tau_{\leqslant 2}R \to \tau_{\leqslant 1}R \to \tau_{\leqslant 0}R$$

is a square-zero extension. Secondly, for any I-adic ordinary ring A, each connection map in the$I$-adic tower of A is a square-zero extension.

Lemma 1.13. *The spherical Witt ring $\mathbb{W}(A)$ can be written as a 2ω-indexed tower limit, with each term identified with a finite sum of shifts $\Sigma^j A$ and each connection map being a square-zero extension.*

Proof. The canonical map $\mathbb{W}(A) \to A$ fits into a pushout square by construction. Therefore by base change we reduce to the case where $A = \mathbb{F}_p$ and $\mathbb{W}(A) = \mathbb{S}_p$. First, one can write $\mathbb{S}_p$ as an ω-indexed limit of its Postnikov tower, with each term consisting of copies of $\Sigma^j\mathbb{Z}_p$ and each connection map being a square-zero extension. Then we can write each copy of $\mathbb{Z}_p$ as an ω-indexed limit of its p-adic tower. One conclude by combining together. □

Remark 1.14. The lemma merely implies that there exists an equivalence between $\mathbb{W}(A)$ and a 2ω-indexed limit of copies of shifts of A. It seems impossible to expect such an equivalence is canonical.

Corollary 1.15. *The spherical Witt functor $\mathbb{W}(-)$ preserves arbituary products.*

1.3. **Properties of the tilt functor.** The tilt functor behaves very crude as one can expect. It can not distingush square-zero extensions, adic elements or the information outside degree zero. More explicitly, we will have the following results

Lemma 1.16. *Let $R \to S$ be a square-zero extension. Then after applying $(-)^\flat$, it becomes an equivalence.*

Proof. Since the map $R^\flat \to S^\flat$ is a map of perfect algebras, it suffices to check the equivalence on the underlying set level. However, the underlying set functor in Perf is corepresented by $\mathbb{F}_p[t^{1/p^\infty}]$. Thus by Example 1.9 and Proposition 1.11, we have equivalences

$$R^\flat \simeq \mathrm{Hom}(\mathbb{F}_p[t^{1/p^\infty}], R^\flat) \simeq \mathrm{Map}(\mathbb{S}_p[t^{1/p^\infty}], R) \simeq \mathrm{Map}(\mathbb{S}_p[t^{1/p^\infty}], S) \simeq S^\flat$$

□

Lemma 1.17. *The tilt functor $(-)^\flat$ sends the span $\pi_0(R) \leftarrow \tau_{\geqslant 0}R \to R$ to equivalences.*

Proof. Almost by definition, since when defining the tilt functor we first take the zeroth homotopy group. □

Lemma 1.18. *Let $x \in \pi_0(R)$ be an element such that $\pi_0(R)$ is x-adic complete, then we have an equivalence $\pi_0(R)^\flat \simeq (\pi_0(R)/x)^\flat$.*

Proof. By completeness, $\pi_0(R)$ can be written as a limit of the x-adic tower of $\pi_0(R)/x$, where the connection maps are square-zero extensions. It follows from Lemma 1.17 and the fact that $(-)^\flat$ preserves limits. □

2. Lubin-Tate theories

The chromatic homotopy theory associates the chromatic height of a cohomology theory with the algebraic height of a formal group. Lubin-Tate theories allow us to attach an $\mathbb{E}_\infty$-ring to universal deformations of formal groups. We first recall some basic facts about formal groups.

2.1. Recollection of formal groups.

Definition 2.1. Let A be an ordinary ring. A **formal group law** over A is a formal series $F \in A[\![t_1, t_2]\!]$ such that

(1) $F(x, y) = F(y, x)$.
(2) $F(x, 0) = F(0, x) = x$.
(3) $F(F(x, y), z) = F(x, F(y, z))$.

We sometimes use $x +_F y$ to denote $F(x, y)$ for simplicity. An isomorphism between two formal group law F, G over A is an invertible formal series $\phi \in A[\![t]\!]$ such that $\phi(x +_F y) = \phi(x) +_G \phi(y)$.

Remark 2.2. The second requirement for a formal group law in Definition 2.1 is important. It forces the group unit to be 0. As a consequence, any formal group law must have forms

$$F(x, y) = x + y + \text{heigher terms}$$

Remark 2.3. Formal groups emerge naturally from cohomology theories. Suppose we have a complex orientable ring spectrum E, by Atiyah-Hirzrebruch spectral sequence $E^*(\mathbb{CP}^\infty) \cong E_*[\![t]\!]$. The homotopy product $m : \mathbb{CP}^\infty \times \mathbb{CP}^\infty \to \mathbb{CP}^\infty$ induces

$$\begin{aligned} m^* : E_*[\![t]\!] &\to E_*[\![t_1, t_2]\!] \\ t &\mapsto F(t_1, t_2) \end{aligned}$$

which determines a formal group law on E^*. Therefore, one has a correspondence

$$\{\text{Complex oriented cohomology theory}\} \to \{\text{Formal group laws}\}$$

Not every formal group law comes from some complex oriented homotopy theory. The obstruction is Landweber exactness. However, we can always reflect a cohomology theory from those universal deformations of formal groups, as we will see later.

Let $\mathcal{FGL} : \mathrm{CRing} \to \mathcal{S}$ be the functor that assigns each ring the groupoid of formal group laws over it. This functor is actually a scheme and is represented by $\mathrm{Spec}(L)$. The ring L can be defined as follows. Let $\mathbb{Z}[a_{ij}]$ be the polynomial ring and choose $F \in \mathbb{Z}[a_{ij}][\![x, y]\!]$ as $F = \Sigma_{i,j} a_{i,j} x^i y^j$. Let $I \in \mathbb{Z}[a_{ij}]$ be the ideal generated by the coeffitients of the relations

(1) $F(x, y) - F(y, x)$.
(2) $F(x, 0) - x$.
(3) $F(F(x, y), z) - F(x, F(y, z))$.

We define $L = \mathbb{Z}[a_{ij}]/I$. One can read from the definition that we must have $\mathcal{FGL}(R) \cong \mathrm{Hom}(L, R)$.

In fact, we have a more manageable description of L

Theorem 2.4 (Lazard)**.** *The ring L is isomorphic to the graded ring $\mathbb{Z}[b_1, \cdots, b_n, \cdots]$ with b_i in degree $2i$.*

We refer to L as the **Lazard ring**. The graded ring structure here is meaningful. In fact, we have the following result.

Theorem 2.5 (Quillen)**.** *Let $\varphi : L \to \mathrm{MU}_*$ be the ring map that classifies the formal group law on MU_* determined by the complex orientation. Then $\varphi : L \xrightarrow{\simeq} \mathrm{MU}_*$ is an isomorphism of graded rings.*

Now we introduce the concept of formal groups, which reform formal group laws into geometric objects.

Definition 2.6. Let A be a ring and F be a formal group law over A. We let $\mathbb{G}_F := \mathrm{Spf} A[\![t]\!]$ be the formal scheme equipped with a multiplication $\mathbb{G}_F \times \mathbb{G}_F \to \mathbb{G}_F$ defined by

$$\begin{aligned} \mu^* : A[\![t]\!] &\to A[\![t_1, t_2]\!] \\ t &\mapsto F(t_1, t_2) \end{aligned}$$

A **formal group** over A is an abelian étale sheaf over A which is locally represented by $\mathbb{G}_F$.

Example 2.7. We use $\mathbb{G}_a$ to denote the formal group $\mathbb{G}_F$ corresponding to the formal group law $F(x, y) = x + y$. We use $\mathbb{G}_m$ to denote the formal group $\mathbb{G}_F$ corresponding to the formal group law $F(x, y) = x + y + xy$.

Now we fix a prime p and let R be an $\mathbb{F}_p$-algebra. Let $\mathbb{G}$ be a formal group over R. The Frobenius on R induces $\mathrm{Frob} : \mathbb{G} \to \mathbb{G}$. We can use this to define the height of a formal group.

Definition 2.8. Let R be an $\mathbb{F}_p$-algebra and let $\mathbb{G}$ be a formal group over R.

(1) We say $\mathbb{G}$ is at least of height n if the multiplication by p on $\mathbb{G}$ factors as

$$\begin{array}{ccc} \mathbb{G} & \xrightarrow{\mathrm{Frob}^n} & \mathbb{G} \\ & \searrow^{\cdot p} & \downarrow{\scriptstyle \phi} \\ & & \mathbb{G} \end{array}$$

(2) We say $\mathbb{G}$ is exactly of height n if it is at least of height n and ϕ is an isomorphism.
(3) We say $\mathbb{G}$ is of infinite height if it is at least of height n for every n.

Remark 2.9. In general, the height does not give a strict stratification of formal groups. That means, there exists some formal group that is at least of height n but is neither at least of height $n + 1$ nor exactly of height n. However, when R is a field, every formal group that is not of infinite height will admits a strict height n. In this case, we will use $\mathrm{ht}(\mathbb{G})$ to denote the height of $\mathbb{G}$.

Remark 2.10. When $R = k$ is a algebraic-closed field, for every height n there exists a unique formal group up to isomorphism.

The height of $\mathbb{G}_F$ can be computed at the level of formal groups. The multiplication by p on $\mathbb{G}_F$ induces a map

$$[p] : \mathrm{Spf}(R[\![t]\!]) \to \mathrm{Spf}(R[\![t]\!])$$

sending t to some formal series. Then $\mathbb{G}_F$ is at least of height n if and only if $[p](t) = h(t^{p^n})$ for some formal series h, $\mathbb{G}_F$ is exactly of height n if and only if $[p](t) = h(t^{p^n})$ and h is invertible.

Example 2.11. In Example 1.9, one can write down the formula of $[p]$ for $\mathbb{G}_a$ and $\mathbb{G}_m$ and after examing the coefficience of the formal series defined by $[p]$, one observes that $\mathrm{ht}(\mathbb{G}_a) = \infty$ and $\mathrm{ht}(\mathbb{G}_m) = 1$.

If we choose v_i to be the coefficience of t^{p^i} in the formal series of $[p]$ for the universal formal group law on MU_*, where each v_i locates in degree $2p^i - 2$, then for any formal group law (R, F), R admits a stratification by

$$(v_0 = p) \subset (v_0, v_1) \subset (v_0, v_1, v_2) \subset \cdots \subset R$$

The quotient $R/(v_0, \cdots, v_n)$ cuts out the $\mathrm{ht} \leqslant n$ part of the formal group law. We define $\mathfrak{m}_n = (v_0, \cdots, v_n)$ and refer to it as the n-**th Landweber ideal** of R.

2.2. **Lubin-Tate functors.** We now introduce the Lubin-Tate functor and compare it with the spherical Witt functor. We begin with some notations.

Definition 2.12. We say an $\mathbb{E}_\infty$-ring R is complex periodic if the followings are satisfied

(1) R is complex orientable.
(2) $\pi_2(R)$ is a rank 1 projective $\pi_0(R)$-module.
(3) The multiplication $\pi_2(R) \otimes_{\pi_0(R)} \pi_n(R) \to \pi_{n+2}(R)$ is an equivalence.

For a complex periodic $\mathbb{E}_\infty$-ring R, we can choose a unit $\mu \in \pi_{-2}(R)$ and press $v_0, v_1, \cdots, v_n$ onto degree 0 by multiplying a power of μ.

Definition 2.13. Let R be an even $\mathbb{E}_\infty$-ring. $\pi_*(R)$ is stratified with Landweber ideals $\mathfrak{m}_i$. One can lift $\pi_*(R)/\mathfrak{m}_n$ to a spectrum $R//\mathfrak{m}_n$, which is defined as

$$R//\mathfrak{m}_n := R/v_0 \otimes \cdots \otimes R/v_n$$

By evenness each R/v_i admits an $\mathbb{E}_1$-structure. Therefore, $R//\mathfrak{m}_n$ gives $\pi_*(R)/\mathfrak{m}_n$ a $\mathbb{E}_1$-lift. We can now introduce the Lubin-Tate functor.

Theorem 2.14 ([Lur17a], Theorem 5.0.2)**.** *Let $\mathcal{FG}^{=n}$ be the category of pairs $(A, \mathbb{G})$ consisting of a perfect $\mathbb{F}_p$-algebra A and a formal group $\mathbb{G}$ of height n. A morphism $\varphi : (A, \mathbb{G}_0) \to (B, \mathbb{G}_1)$ is a pair (f, α), where $f : A \to B$ is a map of rings and $\alpha : f^*\mathbb{G}_0 \xrightarrow{\simeq} \mathbb{G}_1$ is an equivalence. Then there is a functor*

$$E(-;-) : \mathcal{FG}^{=n} \to \mathrm{CAlg}(\mathrm{Sp}_{T(n)})$$

such that

(1) $E(-;-)$ is fully faithful.
(2) For every $(A, \mathbb{G}) \in \mathcal{FG}^{=n}$, $E(A; \mathbb{G})$ is $K(n)$-local and even.
(3) The elements $v_0, \cdots, v_{n-1}$ are regular.
(4) The essential image of $E(-;-)$ consists of $R \in \mathrm{CAlg}(\mathrm{Sp}_{T(n)})$ such that $R//\mathfrak{m}_{n-1}$ is even and $\pi_0(R)/\mathfrak{m}_{n-1}$ is perfect.

(5) *The classical Quillen's formal group over* $\pi_0 E(A;\mathbb{G})$ *is identified with the universal deformation of* $\mathbb{G}$.
(6) *We have an equivalence*

$$\pi_* E(A;\mathbb{G}) \cong W(A)[\![v_1,\cdots,v_{n-1}]\!][\mu^{\pm 1}]$$

where $v_1,\cdots,v_{n-1}$ *locates in degree* 0 *and* μ *is in degree* 2.

Therefore, if we fix a perfect $\mathbb{F}_p$-field k and a given formal group $\mathbb{G}_0$, the functor is simplified to

$$E(-) : \mathrm{Perf}_k \to \mathrm{CAlg}^{\wedge}_{E(k)}$$

which sends $f : k \to A$ to $E(A; f^*\mathbb{G}_0)$.

Remark 2.15. When $A = k$ is a perfect field, then as a spectrum, $E(-,-)$ does not rely on the choice of formal groups, see [LP22].

Now we compare the functor $E(-) : \mathrm{Perf}_k \to \mathrm{CAlg}(\mathrm{Sp}_{T(n)})$ with $\mathbb{W}(-) : \mathrm{Perf} \to \mathrm{CAlg}(\mathrm{Sp}_p)$.

Let $\mathcal{C}$ be a p-complete presentable symmetric monoidal stable ∞-category. The unit functor $\mathbb{S}_p \to \mathcal{C}$ in $\mathrm{CMon}(\mathrm{Pr}^{\mathrm{L}}_{st})$ gives an adjunction

$$\iota^* : \mathrm{CAlg}(\mathrm{Sp}_p) \rightleftharpoons \mathrm{CAlg}(\mathcal{C}) : \iota_*$$

Composing this with the Witt-tilt adjunction, we have

$$\mathbb{W}(-) : \mathrm{Perf} \rightleftharpoons \mathrm{CAlg}(\mathcal{C}) : (-)^\flat$$

where we still use $\mathbb{W}(-)$ and $(-)^\flat$ to denote this pair. The unit $\mathbb{1}_{\mathcal{C}} \in \mathrm{CAlg}(\mathcal{C})$ is sending to some $B \in \mathrm{Perf}$. We obtain an adjunction

$$\mathbb{W}_B(-) : \mathrm{Perf}_B \rightleftharpoons \mathrm{CAlg}(\mathcal{C}) : (-)^\flat$$

which from left to right sends A to $\mathbb{1}_{\mathcal{C}} \otimes_{\iota^*\mathbb{W}(B)} \iota^*\mathbb{W}(A)$.

Now we take $\mathcal{C} = \mathrm{Mod}^{\wedge}_{E(k)}$ to be the category of $T(n)$-local $E(k)$-module. Apply the above construction, we obtain

$$\mathbb{W}_B(-) : \mathrm{Perf}_B \rightleftharpoons \mathrm{CAlg}^{\wedge}_{E(k)} : (-)^\flat$$

for $B = E(k)^\flat$.

Lemma 2.16. *For any* $R \in \mathrm{CAlg}^{\wedge}_{E(k)}$, *we have*

$$R^\flat \simeq \pi_0(R)^\flat \simeq (\pi_0 R/\mathfrak{m}_{n-1})^\flat$$

In particular, we have $E(k)^\flat \simeq k^\flat \simeq k$ *and* $E(-)^\flat \simeq \mathrm{id}$.

Proof. This follows from the discussion in Section 1.3, the fact that ι_* preserves square-zero extensions, and the equivalence between $T(n)$-local and $\mathfrak{m}_n$-complete ([HS99, Proposition 7.10]). □

Therefore, the above adjunction has the form

$$\mathbb{W}_k(-) : \mathrm{Perf}_k \rightleftharpoons \mathrm{CAlg}^{\wedge}_{E(k)} : (-)^\flat$$

Since $E(-)^\flat \simeq \mathrm{id}$, precomposing with the unit map of the adjunction allows us to obtain a natural transformation

$$\mathbb{W}_k(-) \to E(-)$$

One can show that this map is an equivalence by reading the homotopy groups, since we have an equivalence

$$\mathbb{W}_k(A) \simeq \mathbb{W}(A) \otimes_{\mathbb{W}(k)} E(k) \simeq E(A)$$

To conclude, we have the following consequence.

Theorem 2.17. *Fix a perfect field k of character p and a height n, then there is an adjunction*

$$E(-) : \mathrm{Perf}_k \rightleftharpoons \mathrm{CAlg}^{\wedge}_{E(k)} : (-)^\flat$$

such that

1. *$E(-)$ is fully faithful.*
2. *$(-)^\flat \simeq \pi_0(-)^\flat \simeq (\pi_0(-)/\mathfrak{m}_{n-1})^\flat$.*
3. *$\pi_* E(A) \cong W(A)[\![v_1, \cdots, v_{n-1}]\!][\mu^{\pm 1}]$. In particular, $p = v_0, v_1, \cdots, v_{n-1}$ is a regular sequence and $E(A)$ is even.*
4. *The essential image of $E(-)$ consists of $R \in \mathrm{CAlg}^{\wedge}_{E(k)}$ such that $R//\mathfrak{m}_{n-1}$ is even and $\pi_0(R)/\mathfrak{m}_{n-1}$ is perfect.*

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